

WEEK 5

Week 5: Benchmarking and Tuning - Lab Guide

HPC & Cluster Computing

Objective: Characterize the performance of your cluster using standard benchmarks.

Goal: Measure Memory Bandwidth (STREAM) and Compute Performance (HPL), and observe the effect of compiler flags.

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1. The Impact of Compiler Optimizations

Before running complex benchmarks, let's see how much "free" speed we can get just by asking the compiler to work harder.

1.1 Create the Test Code

Create a file named `matrix_test.c`. This code simply multiplies two matrices.

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

#define N 1000

int main() {
```

```
    double *A = malloc(N * N * sizeof(double));
    double *B = malloc(N * N * sizeof(double));
    double *C = malloc(N * N * sizeof(double));

    // Initialize
    for(int i=0; i<N*N; i++) { A[i] = 1.0; B[i] = 2.0; }

    clock_t start = clock();

    // Multiply
    for(int i=0; i<N; i++) {
        for(int j=0; j<N; j++) {
            for(int k=0; k<N; k++) {
                C[i*N+j] += A[i*N+k] * B[k*N+j];
            }
        }
    }

    clock_t end = clock();
    printf("Time: %f seconds\n", (double)(end - start) /
    return 0;
}
```

1.2 Compile and Compare

1. **Compile with No Optimization (-O0)**: `gcc -O0 matrix_test.c -o matrix_slow` `./matrix_slow` *Record the time.*
2. **Compile with Aggressive Optimization (-O3)**: `gcc -O3 matrix_test.c -o matrix_fast` `./matrix_fast` *Record the time.*

Result: You should see a drastic difference (often 10x or more) just from flags.

2. STREAM Benchmark (Memory Bandwidth)

We will use the **STREAM** benchmark to find the peak memory bandwidth of a single node.

2.1 Download and Compile

1. Download the source: `bash wget https://www.cs.virginia.edu/stream/FTP/Code/stream.c`
2. Compile with OpenMP support (to use all memory channels): `bash gcc -O3 -fopenmp -DSTREAM_ARRAY_SIZE=10000000 stream.c -o stream`

2.2 Run the Benchmark

Run it on a single node (e.g., inside an `srun --pty bash` session):

```
export OMP_NUM_THREADS=4   # Match your Core count
./stream
```

Analyze: Look at the "Copy" and "Triad" values. *Result*: `_____ MB/s`

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3. HPL (High Performance Linpack)

Now we measure the Peak Flops of the cluster.

3.1 Install HPL

Building HPL from source is complex (requires BLAS, MPI). For this lab, we will use a standard container or package if available. *Simplified build*. *Assumption*: `hpl` is

3.2 Configure HPL.dat

Create a file named `HPL.dat`. This is the configuration file HPL reads.

```
HPLinpack benchmark input file
Innovative Computing Laboratory, University of Tennessee
HPL.out      output file name (if any)
6            device out (6=stdout,7=stderr,file)
1            # of problems sizes (N)
5000         Ns
1            # of NBs
128          NBs
0            PMAP process mapping (0=Row-,1=Column-major)
1            # of process grids (P x Q)
1            Ps
1            Qs
16.0         threshold
1            # of panel fact
2            PFACTs (0=left, 1=Crout, 2=Right)
1            # of recursive stopping criterium
4            NBMINs (>= 1)
1            # of panels in recursion
2            NDIVs
1            # of recursive panel fact.
1            RFACTs (0=left, 1=Crout, 2=Right)
1            # of broadcast
1            BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
1            # of lookahead depth
1            DEPTHs (>=0)
2            SWAP (0=bin-exch,1=long,2=mix)
64           swapping threshold
0            L1 in (0=transposed,1=no-transposed)
```

```
0            U  in (0=transposed,1=no-transposed)
1            Equilibration (0=no,1=yes)
8            memory alignment in double (> 0)
```

3.3 Run on 1 Node

Run HPL on a single core (Serial):

```
mpirun -np 1 ./xhpl
```

- Look for the final result: `Gflops`.

3.4 Run on 2 Nodes (Parallel)

1. Edit `HPL.dat`: Change `Ps` to `2` and `Qs` to `1` (Total 2 processes).
2. Increase `Ns` to `8000` (More nodes = more RAM = bigger problem).
3. Submit via Slurm: `bash srun -N 2 --ntasks-per-node=1 ./xhpl`

Goal: Did your Gflops double? If not, why? (Network latency, small problem size?)

Submission

Submit a simple report (`Week7_Lab_Report.txt`):

1. **Matrix Time**: -O0 vs -O3 results.
2. **STREAM**: Peak Bandwidth (MB/s).
3. **HPL**: Peak Gflops for 1 Node vs 2 Nodes.

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